HANDS-ON EXERCISES

Setting Up Your Bitcoin Development Environment

★ We will run Bitcoin Core in **regtest** mode

★ Create wallets and generate addresses

★ Mine blocks and send transactions

★ Explore the blockchain with CLI commands

Prerequisite: Bitcoin Core installed

SETUP STEP 1: CREATE DIRECTORY

```
# Create the Bitcoin configuration directory
mkdir -p ~/.bitcoin
```

What this does:

★ Creates a hidden folder `.bitcoin` in your home directory

SETUP STEP 2: CREATE CONFIG FILE

```
# Open the config file for editing
nano ~/.bitcoin/bitcoin.conf
```

Or use any text editor:

★ `code ~/.bitcoin/bitcoin.conf` (VS Code)

★ `vim ~/.bitcoin/bitcoin.conf` (Vim)

SETUP STEP 3: ADD CONFIGURATION

Paste this into bitcoin.conf:

```
# Bitcoin Bootcamp Configuration
regtest=1
server=1
daemon=1
```

SETUP STEP 4: MORE CONFIG

Continue adding to bitcoin.conf:

```
# Allow RPC from localhost
rpcallowip=127.0.0.1
rpcbind=127.0.0.1

# Set fallback fee for regtest
fallbackfee=0.0001

# Index all transactions (needed for Day 2)
txindex=1
```

WHAT DOES THIS CONFIG DO?

Setting	Meaning
`regtest=1`	Use private test network
`server=1`	Enable RPC server
`daemon=1`	Run in background
`rpcuser/password`	Login credentials
`rpcport=18443`	Port for regtest RPC

SETUP STEP 5: START BITCOIN CORE

```
# Start the Bitcoin daemon
bitcoind
```

You should see: (nothing - it runs in background)

```
# Verify it's running
bitcoin-cli -regtest getblockchaininfo
```

SETUP COMPLETE!

If you see this, you're ready:

```
{
  "chain": "regtest",
  "blocks": 0,
  "headers": 0,
  "bestblockhash": "0f9188f13cb7b2c71f2a335e3...",
  ...
}
```

EXERCISE 1: CHECK YOUR NODE

```
# Check blockchain info
bitcoin-cli -regtest getblockchaininfo

# Check network info
bitcoin-cli -regtest getnetworkinfo
```


What do you see?

EXERCISE 2: CREATE WALLETS

```
# Create Alice's wallet
bitcoin-cli -regtest createwallet "alice"

# Create Bob's wallet
bitcoin-cli -regtest createwallet "bob"

# List all wallets
```

EXERCISE 3: GENERATE ADDRESSES

```
# Get address for Alice
bitcoin-cli -regtest -rpcwallet=alice getnewaddress

# Get address for Bob
bitcoin-cli -regtest -rpcwallet=bob getnewaddress

# Check Alice's balance
bitcoin-cli -regtest -rpcwallet=alice getbalance
```

EXERCISE 4: MINE BLOCKS

```
# Store Alice's address in a variable
ALICE=$(bitcoin-cli -regtest -rpcwallet=alice getnewaddress)

# Mine 101 blocks to Alice
bitcoin-cli -regtest generatetoaddress 101 "$ALICE"

# Check balance now
bitcoin-cli -regtest -rpcwallet=alice getbalance
```

Why 101? Coinbase maturity = 100 blocks

Result: Alice now has **50 BTC**!

EXERCISE 5: SEND TRANSACTION

```
# Get Bob's address
BOB=$(bitcoin-cli -regtest -rpcwallet=bob getnewaddress)

# Send 10 BTC from Alice to Bob
bitcoin-cli -regtest -rpcwallet=alice sendtoaddress "$BOB" 10

# Check mempool (unconfirmed transactions)
bitcoin-cli -regtest getmempoolinfo
```

EXERCISE 5: CONFIRM IT

```
# Mine a block to confirm the transaction
bitcoin-cli -regtest generatetoaddress 1 "$ALICE"

# Check Bob's balance
bitcoin-cli -regtest -rpcwallet=bob getbalance

# Mempool should be empty now
bitcoin-cli -regtest getmempoolinfo
```


Result: Bob now has **10 BTC**!

EXERCISE 6: EXPLORE TRANSACTION

```
# List Alice's transactions
bitcoin-cli -regtest -rpcwallet=alice listtransactions

# Get raw transaction (replace TXID with actual ID)
bitcoin-cli -regtest getrawtransaction "TXID" true
```

Look for:

EXERCISE 7: EXPLORE A BLOCK

```
# Get the latest block hash
bitcoin-cli -regtest getbestblockhash

# Get block details (replace HASH)
bitcoin-cli -regtest getblock "HASH" 1

# Get full transaction data
```

EXERCISE 8: ADDRESS TYPES

```
# Legacy (oldest)
bitcoin-cli -regtest -rpcwallet=alice getnewaddress "" "legacy"
# Result: m... or n...

# Native SegWit (recommended)
bitcoin-cli -regtest -rpcwallet=alice getnewaddress "" "bech32"
# Result: bcrt1q...

# Taproot (newest)
bitcoin-cli -regtest -rpcwallet=alice getnewaddress "" "bech32m"
```

WHY SEGWIT MATTERS

Segregated Witness (2017 upgrade)

★

Before: Signatures inside transaction

★ Transaction ID could be changed (malleability)

★ Limited block capacity

★

STOPPING BITCOIN CORE

```
# Stop the Bitcoin daemon gracefully
bitcoin-cli -regtest stop
```

To restart later:

```
bitcoind
```

Your wallets and blockchain data are saved in `~/.bitcoin/regtest/`

DAY 1 SUMMARY

You learned:

★ Configure Bitcoin Core for regtest
★ Create wallets and generate addresses
★ Mine blocks and earn coinbase rewards
★ Send transactions between wallets
★ Explore blocks and transactions via CLI
★ Different address types and SegWit

Tomorrow: Python coding with Bitcoin!

HOMEWORK

★ Send 5 more transactions between Alice and Bob

★ Decode each transaction - understand inputs/outputs

★ Create wallet "charlie" and do 3-way transactions

★ Compare transaction sizes with different address types